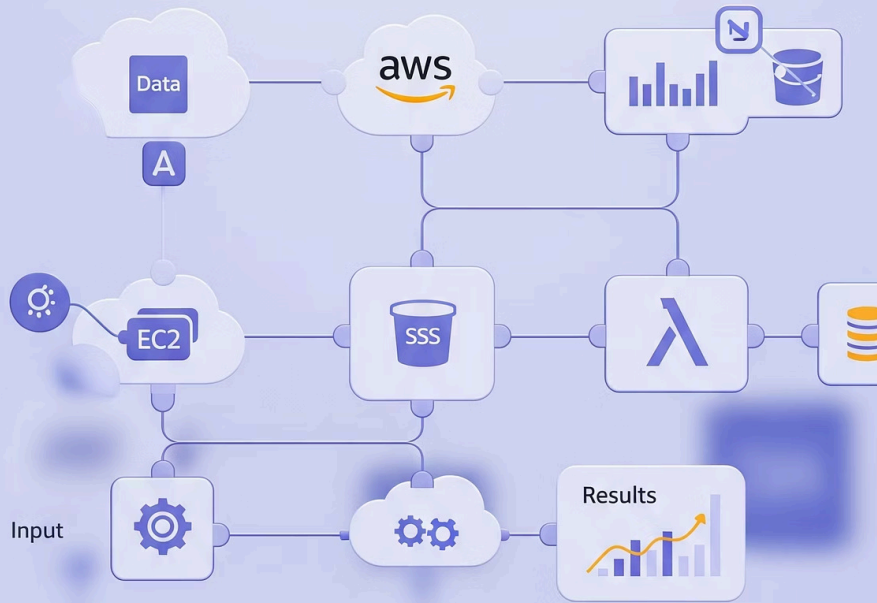




SageMaker Pipelines - Basics

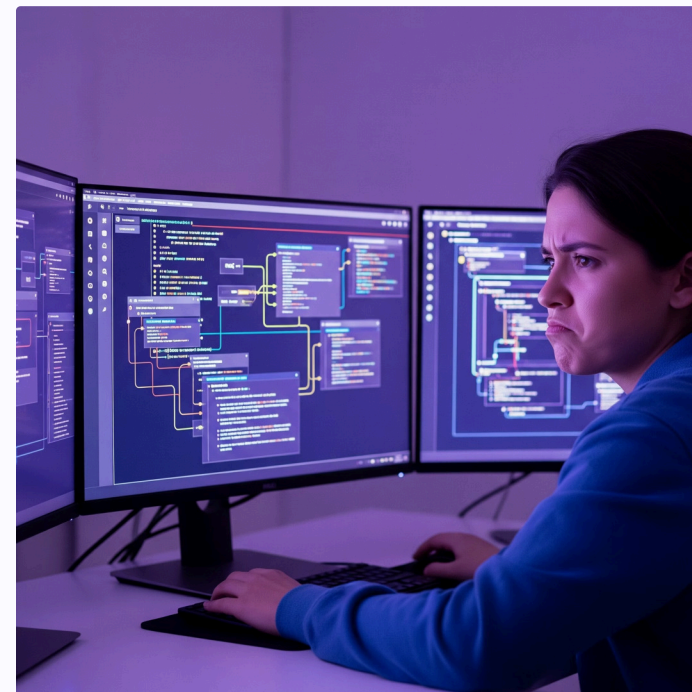
AWS-Native MLOps Orchestration

A streamlined approach to building scalable, reproducible machine learning workflows in Amazon SageMaker.

Why SageMaker Pipelines?

ML lifecycle complexity creates challenges for organizations trying to scale their machine learning operations:

- Manual scripts are **error-prone** & not scalable
- Data preparation, training, and deployment require **consistent orchestration**
- Teams need **end-to-end visibility** across the ML lifecycle



SageMaker Pipelines provides **AWS-native integration** with S3, ECR, CloudWatch, and IAM for a complete MLOps solution.

Core Capabilities

Define ML workflows as pipelines

Create reusable, modular workflows that standardize your ML processes across teams and projects

Built-in step types

Processing, training, tuning, evaluation, and deployment steps with pre-configured container environments

Track lineage

Automatically capture relationships between data, code, and models for compliance and reproducibility

Automate promotions

Implement controlled model progression from development through staging to production environments

Pipeline Workflow Example



Data Preprocessing

Processing job for data cleaning, validation, and transformation



Feature Engineering

Feature Store integration for storing, sharing, and versioning features



Model Training

Estimator step using customizable algorithms on scalable infrastructure



Evaluation

Processing/Eval script to assess model performance



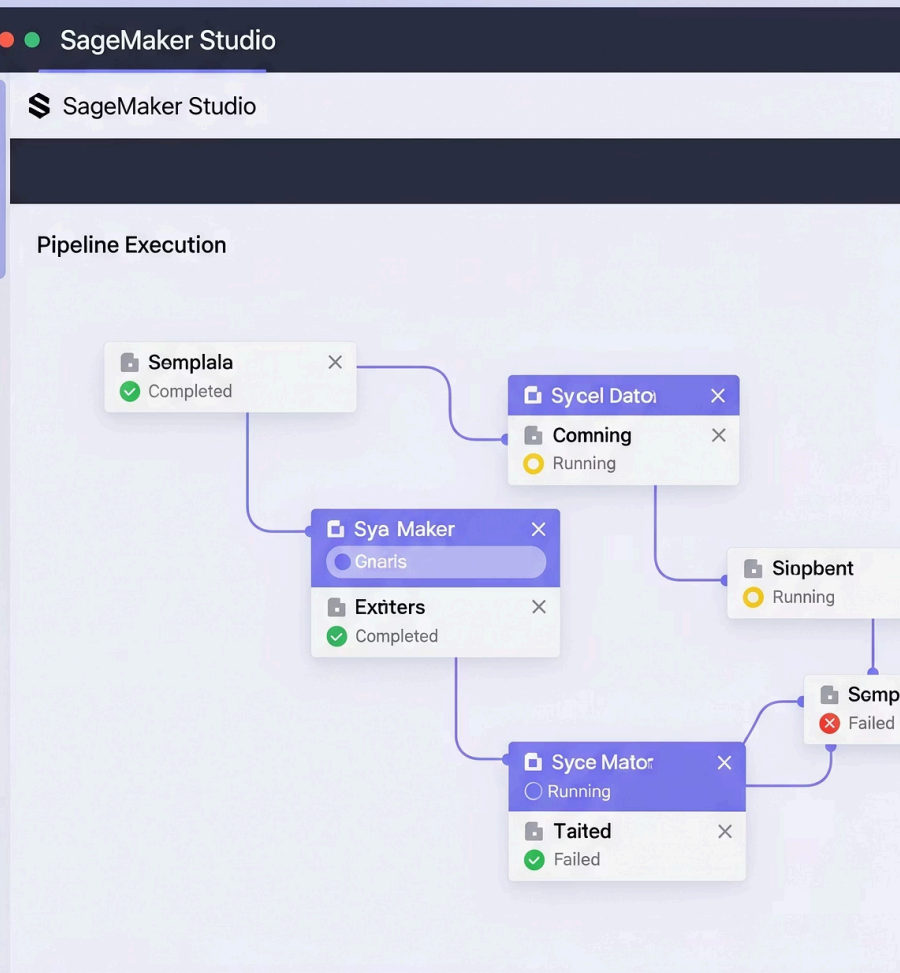
Conditional Step

Promote if accuracy > threshold



Register/Deploy

SageMaker Model Registry + Endpoint deployment



How It Works

Define

Pipelines are defined in **Python SDK** with intuitive syntax that allows teams to create reusable workflow templates

Execute

Each step runs as a containerized job, managed by SageMaker's infrastructure, scaling automatically as needed

Store

Artifacts stored in S3, with complete lineage tracking in SageMaker Experiments for compliance and reproducibility

Monitor

Execution progress and results tracked in **SageMaker Studio UI** with detailed visualizations and logs

Example Code Snippet

```
from sagemaker.workflow.pipeline import Pipeline
from sagemaker.workflow.steps import TrainingStep

train_step = TrainingStep(
    name="TrainModel",
    estimator=estimator,
    inputs={"train": s3_input}
)

pipeline = Pipeline(
    name="MyPipeline",
    steps=[train_step]
)

pipeline.start()
```

This simple example creates a pipeline with a single training step, but real-world pipelines can include dozens of interconnected steps with conditional logic.

Key Components:

- Import SageMaker workflow modules
- Define steps with parameters
- Create pipeline object
- Start execution

The Python SDK abstracts away infrastructure complexity while allowing full customization.



Benefits

100%

Managed
Orchestration

No Airflow/Kubeflow
overhead or infrastructure
to maintain

1000x

Scalability

Train on 1 GPU or scale to
thousands with no code
changes

100%

Reproducibility

Versioned data, code,
models with complete
lineage tracking

IAM

Security

IAM roles, KMS encryption,
private VPC support built-in

Monitoring & Observability



Comprehensive Visibility

- **CloudWatch logs** for each pipeline step
- Track accuracy, loss, and custom metrics
- Complete **lineage view**: datasets → model → endpoint
- Configurable alerts on pipeline failures or drift

SageMaker Pipelines integrates with Amazon CloudWatch to provide centralized logging and monitoring capabilities across your entire ML workflow.

Common Use Cases



Automating Batch Retraining

Schedule periodic model retraining as new data becomes available, ensuring models stay accurate over time without manual intervention



Standardizing ML Workflows

Create consistent, reusable pipeline templates that enforce best practices across data science teams and projects



Model Approval Workflows

Implement conditional deployment with human-in-the-loop approval steps for regulated industries requiring oversight



Enterprise MLOps

Build comprehensive ML platforms on AWS-native infrastructure with integrated security, governance, and scaling

Best Practices

Configuration & Flexibility

- Use **PipelineParameters** for runtime flexibility
- Store all datasets in **S3 with versioning** enabled
- Implement proper error handling and retry logic

Governance & Cost Control

- Integrate with **SageMaker Model Registry** for governance
- Tag resources for cost tracking & auditing
- Implement appropriate instance auto-scaling policies



"Treat your ML pipelines like production software: version control, testing, and continuous improvement are essential."

Key Takeaways

1 AWS-Native ML Orchestration

SageMaker Pipelines provides a purpose-built solution for ML workflow automation without infrastructure overhead

3 Enterprise-Ready Capabilities

Built-in governance, observability, and automation features that scale with your organization's ML initiatives

2 Complete ML Lifecycle Coverage

End-to-end support from data preparation through training, evaluation, deployment, and monitoring

4 AWS-First MLOps

Ideal for organizations building on AWS infrastructure with seamless integration to the broader AWS ecosystem

Ready to build your first SageMaker Pipeline?